

Phaneendra Gavara

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SUMMARY

Data Scientist with experience across statistical modeling, time-series forecasting, and applied machine learning — from SARIMAX demand forecasting to SHAP-explained classification models. M.S. Data Science graduate of Arizona State University (GPA 3.90/4.0).

EDUCATION

Arizona State University <i>M.S. Data Science (Computing & Decision Analytics Concentration)</i> — GPA: 3.90/4.0 • <i>Relevant Coursework:</i> Statistics for Data Analysts, Statistical Machine Learning, Data Mining, Data Visualization & Reporting for IT, Data Science Capstone	Tempe, AZ Aug 2024 – May 2026
IIIT Bhubaneswar <i>B.Tech in Computer Science and Engineering</i> — GPA: 7.14/10	Bhubaneswar, India Dec 2020 – May 2024

TECHNICAL SKILLS

Data Science & Statistics: Pandas, NumPy, scikit-learn, SARIMAX, SHAP, Statistical Modeling, Time-Series Forecasting
ML & Deep Learning: PyTorch, CNNs, DistilBERT, OR-Tools
Data Viz & Analytics: Matplotlib, Seaborn, React Dashboards, Azure SQL, ETL Pipelines, Star Schema Design
Languages & Tools: Python, SQL, R, Git, Docker

PROJECTS

TravelIQ — AI Travel Itinerary Generator <i>Python, Flask, OR-Tools, scikit-learn, SARIMAX, GPT-4o</i> - Built an AI travel planner that generates a complete, feasibility-checked itinerary in under 90 seconds, combining ML recommendations, image generation, and route optimization - Modeled demand and pricing trends with SARIMAX time-series forecasting and scikit-learn to inform itinerary recommendations - Solved multi-stop routing as a Vehicle Routing Problem with Time Windows (VRPTW) using OR-Tools to guarantee feasible schedules	2026
Heart Disease Detection <i>Python, scikit-learn, Pandas, Matplotlib, Seaborn, SHAP</i> - Built a supervised ML classifier achieving 85% accuracy on the Cleveland Heart Disease dataset - Benchmarked multiple model architectures and used SHAP feature analysis to explain individual predictions and surface the strongest risk factors	2024
ASU LLM Evaluation Pipeline <i>Python, LangChain, RAG, RAGAS, GPT-4o, Pinecone, GitHub Actions, React</i> - Built an automated CI/CD evaluation pipeline that scores a university RAG chatbot across 100 test questions on every code push, enforcing 6 automated quality gates before deploy - Cut LLM-judge evaluation cost by 25x and sped up result aggregation by 8x through batched scoring and caching - Integrated RAGAS-based automated scoring with a Pinecone vector store and a GPT-4o judge to quantify RAG quality without manual review	2026
Anomaly Detection in Crowds <i>Python, PyTorch, CNNs, OpenCV, Deep Learning</i> - Built a CNN-based crowd safety system in PyTorch that learns normal crowd behavior from video and flags anomalies in real time - Generated frame-level heatmaps to visualize and localize anomalous activity for rapid human review - Processed video streams with OpenCV for frame extraction and preprocessing ahead of model inference	2025
Shaded Route Planning <i>Python, Image Segmentation, Solar Geometry, NumPy</i> Built a city-scale shade simulation that scores pedestrian walking routes by sun coverage across different times of day	2024

- Combined image segmentation and solar-geometry modeling to estimate shade coverage without requiring live sensor data

InsightHub — AI-Powered Business Analytics Platform

2026

Python, FastAPI, Azure SQL, Azure Blob Storage, Azure OpenAI (GPT-4o), Azure AI Search, React

- Designed and deployed a full-stack Azure analytics platform ingesting 230,000+ rows through a star-schema ETL pipeline into Azure SQL
- Built a FastAPI backend serving business insights to a React dashboard, fully deployed and live in production
- Implemented a GPT-4o-powered RAG search engine over the analytics data so users can query business metrics in natural language

AI Voice Turing Test Model

2024

Python, Transformers, NLP, Audio Processing, Librosa

- Built an audio ML + NLP classifier that distinguishes human from AI-generated calls while analyzing classification behavior for demographic bias, placing 3rd at the ASU Social Bias Hackathon
- Extracted acoustic features with Librosa and fine-tuned a Transformer-based classifier for real-time inference

EXPERIENCE

Visiting Researcher

Aug 2026 – Present

Arizona State University

Tempe, AZ

- Designing, developing, and implementing AI-based approaches and solutions for blockchain technologies
- Enhancing and analyzing machine learning models, algorithms, and intelligent blockchain applications
- Researching and implementing AI techniques to improve blockchain security, scalability, and efficiency

Teaching Assistant & Grader

Aug 2025 – May 2026

Arizona State University, Ira A. Fulton Schools of Engineering (CSE 230 – Computer Architecture)

Tempe, AZ

- Graded student submissions and gave feedback across cache memory calculations, AMAT, MIPS multi-cycle control signals, and pipeline hazards using structured, leniency-calibrated rubrics
- Held office hours and managed course communications (Canvas announcements, student Q&A) for a Computer Architecture course

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (Jul 2026)

AWS Certified AI Practitioner (Jun 2026)

AI Fluency Framework & Foundations – Anthropic (May 2026)

Claude Code 101 – Anthropic (May 2026)

Claude 101 – Anthropic (May 2026)

Introduction to Claude Cowork – Anthropic (May 2026)

LEADERSHIP & AWARDS

Admin Assistant & Executive Member

Jan 2021 – Present

Help N Hope Charitable Society (NGO)

Andhra Pradesh, India

- Co-founded the organization and was appointed Executive Member (2024) for a sustained commitment to making quality education accessible to underprivileged children
- 3rd Place, Social Bias Hackathon – Ethical Spectacle Research x Arizona State University (Oct 2024)